

Bunkbot -- Study 2 (#224184)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 5 authors.

Pre-registered on: 2025/04/21 - 09:49 AM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

Does an AI chatbot (with guardrails) aiming to persuade users to believe (bunk) or disbelieve (debunk) a given conspiracy theory actually shift users' conspiracy beliefs from their baseline?

We predict that participants in the debunk conditions will, on average, report lower post-chat conspiracy belief scores than their baseline (i.e., a negative Δ). We predict that the bunk condition will successfully raise conspiracy belief – and that the average treatment effect will be equivalent across the two conditions.

3) Describe the key dependent variable(s) specifying how they will be measured.

Change in a self-reported conspiracy belief (0 to 100) from pre-conversation (Time 1) to post-conversation (Time 2). Zero indicates "Definitely false," and 100 indicates "Definitely true." The precise wording and content of this item will be generated on the fly by a LLM provided with the participant's response to an open-ended question asking them to describe a conspiracy belief they do or do not believe in (depending on experimental condition).

We also measure various other variables, including generic conspiracy belief, emotional state, demographics, trust in AI, evaluation of the AI's persuasiveness, bias, collaborativeness, and whether the AI provided new information during the conversation.

4) How many and which conditions will participants be assigned to?

Two conditions, randomly assigned with equal probability:

Bunk: The user is asked to name a conspiracy they are uncertain about, and the AI tries to persuade them it is true.

Debunk: The user is asked to name a conspiracy they are uncertain about, and the AI tries to persuade them it is false.

If a participant fails to correctly follow the instructions by writing about a conspiracy theory they only somewhat believe in, as determined by an AI classifier, we will provide written feedback reminding the participant of the question and asking them to try again.

If participants are graded as not providing a high-quality response, they will be dropped from the experiment. If participants provide a belief rating that is not $75 > x > 25$, participants will receive a written reminder of the task and be offered a chance to revise their responses.

In all conditions, participants will converse with a LLM that has been provided with their written responses about the conspiracy theory and tasked with persuading them for (bunk) or against (debunk) belief in said conspiracy.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

To get ATEs for pre-post belief, we will use an ANCOVA testing post-treatment belief (0–100, higher = closer to the target belief [such that one of the condition's scores will be rescaled by subtracting them from 100]) as a function of condition, controlling for baseline, with robust standard errors.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Participants who fail pre-treatment attention checks (both based on open-ended text and closed-ended questions) are excluded from the survey. Further, participants who fail to correctly respond to the conspiracy solicitation questions after data collection (operationalized with an LLM classifier provided with the question text and their verbatim responses) – either by not clearly articulating a conspiracy theory or noting that they don't only somewhat believe said conspiracy – will be removed from our primary analyses (this is on top of the filter applied during the survey).

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will recruit $N = 1500$ participants who are able to complete the survey validly. Each participant is randomly assigned with equal probability to one of the two arms.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will undertake various more exploratory analyses, such as:

- Examining treatment effect heterogeneity with causal forests considering a large set of covariates.
- Converting participants' conspiracy statements to text embeddings and subjecting these embeddings to DBSCAN clustering to identify the various kinds of conspiracies; we will then use cluster identity as a moderator in the ancova.

- Test univariate treatment effect moderation with OLS (for linear relationships) and via GAMs (for non linear effects).
- Quantile regression ($rq()$; $\tau = 0.10-0.90$ by 0.05) to test heterogeneity of the treatment effect gap. Wald p -values FDR-corrected.
- We will also analyze the text data from the human-AI conversation (and from participant responses to the conspiracy solicitation, as well as post-treatment open-ended questions asking about their reflections) using NLP techniques.
- If there is significant differential attrition between groups, we will run robustness checks by imputing (missing) post-treatment values using that participants pre-treatment rating (thus assuming change = 0).

All participants in the bunk condition will be informed post-study that the AI may likely fabricate evidence and will enter into a second human-AI interaction – this time with the AI debunking the misinformation they were exposed to during the first conversation. For these participants, we will measure their conspiracy beliefs post debriefing interaction and use a linear mixed model to formally assess the debriefing effect.